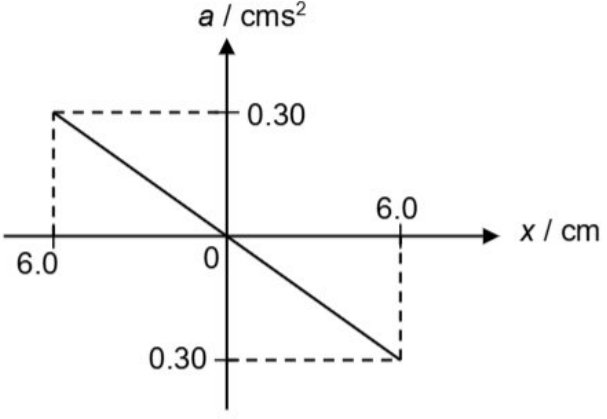
12 A particle moves such that its acceleration a is related to its displacement x from a fixed point as shown.



What is its angular frequency?

A 0.050 rad s^{-1} **B** 0.22 rad s^{-1} **C** 4.5 rad s^{-1} **D** 20 rad s^{-1}